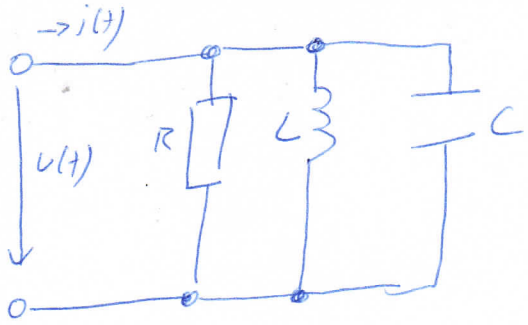
tkysela



$R = 500\,\Omega$

$L = 2\,H$

$C = 0{,}5\,uF$

$I_0 = 90{,}4\,mA$

$I_{m_1} = 60\,mA$

$I_{m_3} = 8{,}435\,mA$

$\omega = 1000\,rad\,s^{-1}$

$\varphi_3 = -0{,}624\,rad$

$$i(t) = I_0 + I_{m_1}\sin(\omega t) + I_{m_3}\sin(3\omega t + \varphi_3)$$

$\hat{I}_0 = 0{,}0904\ A$

$\hat{I}_1 = 0{,}06\ A$

$\hat{I}_3 = 0{,}008435\, e^{-j\,0{,}624}\ A$

$U_0 = 0{,}0904 \cdot 500 = 45\,[V]$

$\hat{U}_1 = 0{,}06 \cdot 500 = 30\,[V]$

$\hat{U}_3 = 4{,}2175\, e^{-j\,0{,}624}\,[V]$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C = \frac{1}{500} + \frac{1}{j1000 \cdot 2} + j \cdot 1000 \cdot 0{,}5 \cdot 10^{-6} = \frac{1}{500}$$

$Z = 500$

$$\boxed{u(t) = 45 + 30\sin(1000t) + 4{,}2175\sin(3000t - 0{,}624)}$$

$$\boxed{I_{ef} = \sqrt{I_0^2 + \frac{1}{2}\left(I_{m_1}^2 + I_{m_3}^2\right)} = 0{,}1\,[A]}$$